

Introduction to Modeling for Formal Verification

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Today's goals



▼

Module 1

Introduction to Model Checking

▼

Module 2

Modeling

▼

Module 3

Modeling System Composition

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- Modeling Concurrency Using Transition Systems
- ✓

Concurrency and Interleaving

Video • 10 min
- ✓

Concurrency and Interleaving: Examples

Video • 8 min
- ✓

Principles of Model Checking - Section 2.2.1, (Page 35-39)

Reading • 30 min
- ✓

Modeling Concurrency Using Transition Systems Quiz

Graded Assignment • Grade: 100%

- Modeling Communications Via Shared Variables
- ✓

Interleaving Program Graphs

Video • 11 min
- ✓

Critical Actions and Atomicity

Video • 10 min
- ✓

Peterson's Mutual Exclusion Algorithm

Video • 12 min
- ✓

Principles of Model Checking - Section 2.2.2, (Page 39-47)

Reading • 30 min
- ✓

Modeling Communications Via Shared Variables Quiz

Graded Assignment • Grade: 100%

- Modeling Communications Via Handshaking
- ✓

Synchronization and Handshaking

Video • 12 min
- ✓

Example: Railroad Crossing

Video • 13 min
- ✓

Principles of Model Checking - Section 2.2.3, (Page 48-52)

Reading • 30 min
- Modeling Communications Via Handshaking Quiz

Graded Assignment • 10 min

- Synchronous Parallelism
- Synchronous Product - Part 1

Video • 10 min
- Synchronous Product - Part 2

Video • 10 min
- The State Space Explosion Problem

Video • 9 min
- Principles of Model Checking - Section 2.2.6, (Page 73-75)

Reading • 20 min
- Synchronous Parallelism Quiz

Graded Assignment • 20 min

Principles of Model Checking - Section 2.2.3, (Page 48-52)

 0:13 / 0:50

2x





Textbook - Principles of Model Checking - Chapter 2.2.3 (page 48-52)

Section 2.2.3 introduces another communication model via handshake. This communication model is often used to describe distributed systems where the consisting processes run on compute nodes without sharing memory. In this model, communication is conducted via passing messages among the processes. Handshake is a particular type of such a communication model where message passing between two processes are abstracted as shared actions. This section explains how to find the transition system model for the system by composing the transition system models for the consisting processes that share actions representing message passing. It also shows examples illustrating the composition via handshake.

Go to next item

✓ Completed

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